

# cse 242 hw1

Ian Chi

April 10, 2026

## 1

1. (a) the characteristic polynomial for this matrix  $\Sigma$  is

$$\lambda^2 - 41\lambda + 175 \quad (1)$$

which results in eigenvalues of

$$\frac{1}{2}(41 \pm 3\sqrt{109}) \quad (2)$$

setting  $(\Sigma - \lambda I)v = 0$ , solving for  $v$  gives the eigenvectors of  $\lambda$ . this results in  $(\frac{1}{10}(3 \pm \sqrt{109}), 1)$

we can check if

$$\begin{pmatrix} 25 & 15 \\ 15 & 16 \end{pmatrix} \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} = \frac{1}{2}(41 \pm 3\sqrt{109}) \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} \quad (3)$$

and we find that this equation holds.

$$\begin{aligned} \begin{pmatrix} 25 & 15 \\ 15 & 16 \end{pmatrix} \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} &= \begin{pmatrix} 15 + \frac{5}{2}(3 \pm \sqrt{109}) \\ 16 + \frac{3}{2}(3 \pm \sqrt{109}) \end{pmatrix} \\ &= \frac{1}{2}(41 \pm 3\sqrt{109}) \begin{pmatrix} \frac{1}{10}(3 \pm \sqrt{109}) \\ 1 \end{pmatrix} \end{aligned} \quad (4)$$

- (b) a plot of the pdf would have 1 std extend 4 units in 1 axis and 5 in the other. the level curve would be an ellipse with some positive correlation between the two axis since the cov is positive.
- (c) the direction that preserves the most variance is the one corresponding to the largest eigenvector of the matrix.  
the fraction of variance preserved would be  $\lambda/41 \approx 0.88$  where 41 is the trace of the cov matrix.

2.

$$\begin{aligned}
Tr(AB) &= \sum_i^m (AB)_{ii} \\
&= \sum_i^m \sum_j^n (A_{ij} B_{ji}) \\
&= \sum_j^n \sum_i^m (A_{ij} B_{ji}) \\
&= \sum_j^n (BA)_{jj} \\
&= Tr(BA)
\end{aligned} \tag{5}$$

3.

$$\begin{aligned}
E[x^T A x] &= E[(\mu + (x - \mu))^T A (\mu + (x - \mu))] \\
&= E[(\mu^T A \mu) + (\mu^T A (x - \mu)) \\
&\quad + ((x - \mu)^T A \mu) + ((x - \mu)^T A ((x - \mu)))] \\
&= (\mu^T A \mu) + E[(\mu^T A (x - \mu)) \\
&\quad + ((x - \mu)^T A \mu) + ((x - \mu)^T A ((x - \mu)))] \\
&= (\mu^T A \mu) + E[(\mu^T A (x - \mu)) \\
&\quad + ((x - \mu)^T A \mu) + ((x - \mu)^T A ((x - \mu)))] \\
&= (\mu^T A \mu) + E[(\mu^T A (x - \mu)) \\
&\quad + ((x - \mu)^T A \mu) + ((x - \mu)^T A ((x - \mu)))] \\
&= (\mu^T A \mu) + E[(\mu^T A (x - \mu)) \\
&\quad + ((x - \mu)^T A \mu) + ((x - \mu)^T A ((x - \mu)))] \\
&= (\mu^T A \mu) + E[A((x - \mu)^T ((x - \mu)))] \\
&= (\mu^T A \mu) + tr(A\epsilon)
\end{aligned} \tag{6}$$

4.

$$E[X] = \int_{-1}^1 \frac{x}{2} dx = 0 \tag{7}$$

$$E[X^2] = \int_{-1}^1 \frac{x^2}{2} dx = \frac{1}{3} \tag{8}$$

$$E[X^3] = \int_{-1}^1 \frac{x^3}{2} dx = 0 \tag{9}$$

$$E[XY] = \int_{-1}^1 \frac{x^3}{2} dx = 0 \tag{10}$$

Cov is however,

$$E[X^3] - E[X]E[X^2] = 0 \tag{11}$$

since cov is zero, so is correlation but XY are obviously dependent.

## 2

1. (a)

$$P(d|+) = \frac{P(+|d)P(d)}{P(+)} = P(d|+) = \frac{P(+|d)P(d)}{P(+|d) \cdot P(d) + P(+|\neg d) \cdot P(\neg d)}$$

$$= \frac{1}{100} \quad (12)$$

$$P(+|d) \cdot P(d) = 0.99 \cdot 0.0001 = 0.000099$$

$$P(+) = (0.99 \cdot 0.0001) + (0.01 \cdot 0.9999)$$

$$P(D|+) = \frac{0.000099}{0.010098} \approx 0.0098$$

- (b) this statement doesnt work because it doesnt take in account of the prior probability of having the disease.

- (c)

$$P(d|+) = \frac{P(+|d) \cdot P(d)_{new}}{P(+)_{new}}$$

$$0.99 \cdot 0.0098039 \approx 0.0097059$$

$$P(+) = (0.99 \cdot 0.0098039) + (0.01 \cdot 0.9901961) \approx 0.0196$$

$$P(d|+_{second}) = \frac{0.0097059}{0.0196079} \approx 0.495$$

- (d)

$$\frac{P(+|d) \cdot P(d)}{P(+|d) \cdot P(d) + (1-s) \cdot P(\neg d)} \geq 0.5$$

$$s \geq 0.99990099...$$

2. (a) the prosecutor is miscalculating the prior. since the prior is known, the statement that the chance is exactly .99 of being guilty is wrong
- (b) the defender is not acknowledging that having this evidence changes the prior. future evidence will build on this new prior instead of the old one.
- (c) you would need to know the prior probability of the defendant being guilty.

3. (a) maximize

$$P^3((1-P)^2) \quad (13)$$

we get the max at .6

- (b) instead we maximize

$$P^3((1-P)^2)2P \quad (14)$$

which gives  $P = \frac{2}{3}$

- (c)  $a = b = 1$  means there were two flips, one heads one tails. when they equal two, there were 4 flips w 2 heads 2 tails. as  $a + b$  gets large, it means that there were many flips before and that you are very certain to the probability of heads.  $\hat{\theta}_{MAP}$  therefore does not change much given an extra sample point or two.

4.

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$L(\mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right)$$

we can take log to make derivatives easier while preserving maximums.

$$l(\mu, \sigma^2) = \ln \left[ (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right) \right]$$

solving for  $\mu$ , we can take  $\sigma^2$  to be constant,

$$\frac{\partial l}{\partial \mu} = \frac{\partial}{\partial \mu} \left( -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right) = \frac{1}{\sigma^2} \left( \sum_{i=1}^n x_i - n\mu \right) = 0$$

gives  $\mu = 4$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2} \frac{1}{\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2$$

implies

$$-n\sigma^2 + \sum_{i=1}^n (x_i - \mu)^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

$$\hat{\sigma}^2 = \frac{(1-4)^2 + (3-4)^2 + (5-4)^2 + (7-4)^2}{4} = 5$$

5. (a) Likelihood:  $P(x_1|\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_1-\mu)^2}{2\sigma^2}\right)$   
 Prior:  $P(\mu) = \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{(\mu-\mu_0)^2}{2\tau^2}\right)$  where  $\mu_0 = 0$  and  $\tau^2 = 1$ .

$$\log P(\mu|x_1) \propto -\frac{(5-\mu)^2}{2(4)} - \frac{\mu^2}{2(1)}$$

$$\frac{d}{d\mu} \left[ -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2} \right] = \frac{5-\mu}{4} - \mu = 0$$

this gives  $\hat{\mu} = 1$

(b) we instead get

$$\log P(\mu|X) \propto - \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\tau^2}$$

$$\frac{d}{dx} \log P(\mu|X) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) - \frac{\mu - \mu_0}{\tau^2} = 0$$

we get  $\hat{\mu} = \frac{n}{n+4} \bar{x}$  where  $\bar{x}$  represents the mean of the samples.

(c) the first term in the  $\hat{\mu}$  in part b goes to 1 which means it converges to the mean of  $x$  as  $n \rightarrow \infty$ .

6. (a)

$$\begin{aligned} \text{Var}(X) &= \frac{\sum (x - \mu)^2}{N} \\ &= \frac{\sum x^2 - 2x\mu + \mu^2}{N} \\ &= \frac{\sum x^2 - \sum 2x\mu + \mu^2}{N} \\ &= \frac{\sum x^2}{N} - \frac{2\mu \sum x}{N} + \mu^2 \\ &= \frac{\sum x^2}{N} - 2\mu\mu + \mu^2 \\ &= E[X^2] - E[X]^2 \end{aligned} \tag{15}$$

(b)

$$\begin{aligned} \text{Var}(aX + b) &= E[(aX + b)^2] - E[aX + b]^2 \\ &= E[(a^2X^2 + 2aXb + b^2)] - (aE[X] + b)^2 \\ &= E[a^2X^2 + 2aXb + b^2] \\ &\quad - [(aE[X])^2 + 2aE[X]b + b^2] \\ &= E[a^2X^2] - (aE[X])^2 \\ &= a^2\text{Var}(X) \end{aligned} \tag{16}$$

(c)

(d)

$$Var(X + Y) = E[(X + Y)^2] - E[X + Y]^2 \quad (17)$$

$$= E[X^2 + 2XY + Y^2] - (E[X] + E[Y])^2 \quad (18)$$

$$= E[X^2 + 2XY + Y^2] - (E[X] + E[Y])^2 \quad (19)$$

$$= E[X^2 + 2XY + Y^2] \quad (20)$$

$$- (E[X]^2 + 2E[X]E[Y] + E[Y]^2)$$

$$= E[X^2] + E[2XY] + E[Y^2] \quad (21)$$

$$- (E[X]^2 + 2E[X]E[Y] + E[Y]^2)$$

$$= E[X^2] + 2E[X]E[Y] + E[Y^2] \quad (22)$$

$$- (E[X]^2 + 2E[X]E[Y] + E[Y]^2)$$

$$= E[X^2] + E[Y^2] - (E[X]^2 + E[Y]^2) \quad (23)$$

$$= Var(X) + Var(Y) \quad (24)$$

note the lines 21 and 22 are the lines that require independence.  
 $E[XY] = E[X]E[Y]$